

Programming for Business

LO2 & LO3

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FAST-NUCES | Semester 2

Class Objectives

- By the end of this class, you will be able to:
- • Understand computational thinking in business
- • Explain input, processing, and output clearly
- • Write simple pseudocode for business problems
- • Choose correct data structures for business data

What is Computational Thinking?

- Computational thinking is a structured approach to solving problems.
- It involves:
 - • Breaking problems into parts
 - • Identifying patterns
 - • Focusing on relevant details
 - • Creating step-by-step solutions

Why Computational Thinking is Important in Business

- Businesses use computational thinking to:
 - • Make better decisions
 - • Reduce errors
 - • Automate repetitive tasks
 - • Improve efficiency

Four Pillars of Computational Thinking

- The four main pillars are:
- 1. Decomposition
- 2. Pattern Recognition
- 3. Abstraction
- 4. Algorithm Design

Decomposition – Definition

- Decomposition means breaking a large problem into smaller, manageable parts.
- This makes complex problems easier to solve.

Decomposition – Business Example

- Example: Running a retail store
 - • Sales management
 - • Inventory control
 - • Staff management
 - • Financial accounting

Pattern Recognition – Definition

- Pattern recognition means identifying similarities or trends in data over time.
- It helps businesses predict future outcomes.

Pattern Recognition – Business Example

- Example:
 - • Monthly sales trends
 - • Seasonal demand for products
 - • Repeat customer behavior

Abstraction – Definition

- Abstraction means focusing only on important information and ignoring unnecessary details.
- It simplifies decision-making.

Abstraction – Business Example

- Profit Calculation:
- Profit = Total Sales – Total Cost
- Only relevant data is considered.

Algorithm Design – Definition

- An algorithm is a clear, step-by-step method to solve a problem.
- Algorithms ensure consistency and accuracy.

Algorithm Design – Business Example

- Salary Calculation Steps:
- 1. Input basic salary
- 2. Add allowances
- 3. Deduct tax
- 4. Display final salary

Input, Processing, and Output (IPO Model)

- The IPO model describes how systems work:
 - • Input: Data provided to the system
 - • Processing: Calculations or decisions
 - • Output: Final result

IPO – Business Example

- Example: Online Order System
- Input: Order details
- Processing: Price calculation and verification
- Output: Order confirmation

IPO – In-Class Activity

- Activity:
- ATM Cash Withdrawal
- Identify the Input, Processing, and Output.

What is Pseudocode?

- Pseudocode is a simple way of writing logic in plain English.
- It helps plan solutions before writing actual programs.

Pseudocode – Business Example

- START
- Input Sales
- Input Cost
- Profit = Sales – Cost
- Display Profit
- END

Pseudocode – Real-Life Scenario

- Scenario: Giving discount to customers
- If total purchase exceeds a limit, give discount.

Variables – Definition

- A variable is a named storage location that holds data which can change.
- Variables store business values such as sales and expenses.

Variables – Business Examples

- Examples of variables:
 - • Sales
 - • Cost
 - • Profit
 - • Tax rate

Operators – Definition

- Operators are symbols used to perform calculations.
- Common operators include addition, subtraction, multiplication, and division.

Operators – Business Example

- Tax Calculation Example:
- $\text{Tax} = \text{Profit} \times \text{Tax Rate}$

Lists – Definition

- A list is a collection of multiple values stored together.
- Lists are used when data items are similar.

Lists – Business Example

- Daily Sales for a Week:
- [5000, 7000, 6500, 8000, 7200]

Processing Lists – Business Use

- From a sales list, businesses can:
 - • Calculate total sales
 - • Find average sales
 - • Identify best sales day

Tuples – Definition

- A tuple is a collection of values that should not be changed.
- It is used for fixed records.

Tuples – Business Example

- Invoice Record:
- (Invoice Number, Date, Total Amount)

Dictionaries – Definition

- A dictionary stores data in key-value pairs.
- Each key is linked to a specific value.

Dictionaries – Business Example

- Employee Salary Record:
- Employee ID → Salary

Sets – Definition

- A set stores unique values only.
- Duplicate values are automatically removed.

Sets – Business Example

- Unique Customers Visiting a Store

Integrated Practice Scenario

- A company tracks:
 - • Daily sales
 - • Employee salaries
 - • Unique customers
- Choose appropriate data structures.

Solution – Practice Scenario

- • Daily sales → List
- • Employee salaries → Dictionary
- • Unique customers → Set